Lösung: Cramer'sche Regel

a)
$$x_1 + 2x_2 = 3$$
$$4x_1 + 6x_2 = 8$$

$$D = \det\begin{pmatrix} 1 & 2 \\ 4 & 6 \end{pmatrix} = 6 - 8 = -2$$

$$D_1 = \det\begin{pmatrix} 3 & 2 \\ 8 & 6 \end{pmatrix} = 18 - 16 = 2$$

$$D_2 = \det\begin{pmatrix} 1 & 3 \\ 4 & 8 \end{pmatrix} = 8 - 12 = -4$$

$$x_1 = \frac{D_1}{D} = \frac{-2}{2} = -1 \quad ; \quad x_2 = \frac{D_2}{D} = \frac{-4}{-2} = 2$$

$$\mathbb{L} = \{(x_1 = -1; x_2 = 2)\}$$

b)
$$x + y + z = 6$$
$$2x - 2y + z = 2$$
$$3x - 2y + z = 2$$

$$D = \det\begin{pmatrix} 1 & 1 & 1 \\ 2 & -2 & 1 \\ 3 & -2 & 1 \end{pmatrix} = 1\cdot(-2)1 + 1\cdot 1\cdot 3 + 1\cdot 2\cdot(-2) - 1\,(-2)\cdot 3 - 1\cdot 1\cdot(-2) - 1\cdot 2\cdot 1$$
$$= -2 + 3 - 4 + 6 + 2 - 2 = 3$$

$$D_x = \det\begin{pmatrix} 6 & 1 & 1 \\ 2 & -2 & 1 \\ 2 & -2 & 1 \end{pmatrix} = 6\cdot(-2)1 + 1\cdot 1\cdot 2 + 1\cdot 2\cdot(-2) - 1\,(-2)\cdot 2 - 6\cdot 1\cdot(-2) - 1\cdot 2\cdot 1$$
$$= -12 + 2 - 4 + 4 + 12 - 2 = 0$$

$$D_y = \det\begin{pmatrix} 1 & 6 & 1 \\ 2 & 2 & 1 \\ 3 & 2 & 1 \end{pmatrix} = 1\cdot 2\cdot 1 + 6\cdot 1\cdot 3 + 1\cdot 2\cdot 2 - 1\cdot 2\cdot 3 - 6\cdot 2\cdot 1 - 1\cdot 1\cdot 2$$

$$= 2 + 18 + 4 - 6 - 12 - 2 = 4$$

$$D_z = \det\begin{pmatrix} 1 & 1 & 6 \\ 2 & -2 & 2 \\ 3 & -2 & 2 \end{pmatrix} = 1\cdot(-2)\cdot 2 + 1\cdot 2\cdot 3 + 6\cdot 2\cdot(-2) - 6\cdot(-2)\cdot 3 - 1\cdot 2(-2) - 1\cdot 2\cdot 2$$

$$= -4 + 6 - 24 + 36 + 4 - 4 = 14$$

$$x = \frac{D_x}{D} = \frac{0}{3} = 0\,; \quad y = \frac{D_y}{D} = \frac{4}{3}\,; \quad z = \frac{D_z}{D} = \frac{14}{3}\,;$$

Lösungsmenge: $\mathbb{L} = \{(x=0;\ y=4/3;\ z=14/3)\}$